FEATURES

# Four-day Work Week Improves Environment

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Abstract In the United States all levels of government face pressure to downsize. The costs associated with government services have continued to outpace revenue sources. Citizens are demanding that government provide better levels of service at lower costs.

In an effort to increase efficiency without increasing staff, the Lake County Health Department, Lake County Illinois, experimented with using a flexible four-day work week. The initiative was designed to increase staff field time while reducing automobile emissions in a cooperative effort to meet air quality targets by the U.S Environmental Protection Agency (EPA).

## Introduction

The use of flexible scheduling has increased in the United States and other industrialized nations. "Evidence shows people are more productive when they are given flexibility" (1).

The Environmental Health Services (EHS) unit, Lake County Illinois, has experimented with a four-day work week to increase efficiency. The program was initially designed to increase staff field time while reducing automobile emissions. It was one way to help meet EPA's auto emission reduction initiative, which has since been withdrawn.

## Methods

EHS evaluated the four-day work week using one of its satellite offices as an experimental work group. The purpose of the four-day work week was sixfold. First, it was designed to increase the sanitarians' field time. A four-day work week could reduce travel time from work in the field to the office. Theoretically, staff would spend more time in the field each day because of the longer work hours and fewer return trips to the office.

The second feature was to reduce employee automobile use and the number of employee automobiles in the parking lot by 20 percent. The environment would benefit by the reduction of automobile emissions and fuel conservation. For example, 50 trips are made each week if all 10 employees drive to the office five days a week. In addition, 10 cars would be in the parking lot five days a week. If all 10 employees work a four-day work week, the number of trips to the office from home is reduced to 40. Thus, there is a 20 percent reduction in both the number of trips to the office and in the number of automobiles in the parking lot each week. Fewer trips to the office from home also decreased the employees' transportation costs.

The third objective was to reduce travel expense. Fewer trips from the office to the field would save mileage expense. Unproductive mileage back to the office at the end of each day would be reduced.

A fourth goal was to increase the public's and contractors' access to EHA by increasing the number of hours of operation. The previous office hours were 8:30 a.m. to 5:00 p.m. Hours of operation were expanded to 7:30 a.m. to 5:30 p.m., representing an 18 percent increase. "Sometimes relatively simple flextime arrangements can improve service levels to customers" (2).

A fifth purpose was to allow the employees to have an extra day off each week. This would give them greater access to community services resulting in less use of sick leave and personal leave. Scheduled medical appointments or personal business could be done on staff's extra day off.

The final objective was designed to help employees with small children reduce their daycare cost. Employees could spend an extra day with their children and reduce their day care expenses one day per week.

The six objectives above were chosen because they would be easy to measure and would remain consistent despite any variations to work loads. They also represent items important to both the health department and the employees.

## Four-day Work Week Design

The Wauconda Satellite Office business hours were changed to 7:30 a.m. to 5:30 p.m., Monday through Friday. The work week consisted of one nine-hour day and three 9.5 hour days, making a 37.5 hour week. Staff schedules were staggered to assure proper office coverage (Table 1).

Staff field time, mileage, and leave balances were tracked by the health department's AS 400 mini computer. These values were compared with targeted objectives noted in the methods section of this article.

Since staff members are regularly out of the office three days per week, work teams were set to assure appropriate coverage. It would be unacceptable to tell clients that their work would have to sit until a certain sanitarian was back in the office after a three-day weekend. To prevent these problems, each township was assigned a back-up sanitarian to perform duties when the area sanitarian was not working. Each team member would have different days off.

To counteract potential communication problems, mandatory staff meetings were held every Tuesday. An agenda was posted prior to the meetings, allowing time for staff to prepare. It also provided them the opportunity to add any issues they wished to discuss. Part of each staff meeting was designated for problem solving. For example, if a staff member were having difficulty meeting a deadline, the entire team would discuss and devise a solution. One work team found it necessary to balance the food service workload equally among all staff members.

One of the most complicated issues was the process of crediting EHS employees with holiday pay. After extensive research, it was decided that employees working flexible schedules should receive hour-for-hour credit compensation. For example, if a holiday fell on Monday, an employee who was scheduled to work 9.5 hours would be compensated 9.5 hours. A complication arose because individuals working a traditional five-day work schedule would only be compensated 7.5 hours of pay for the same day off. Compensating employees for the hours that they were scheduled to work was a fair solution to the problem.

## Findings

The first goal of the four-day work week was to increase field time as a percentage of the sanitarian's day. The objective was achieved, though not to the degree expected. Sanitarian field time increased from 24 percent to 26 percent of total time (Tables 2 and 3). The modest two percent increase, while positive,

### TABLE 1

**New Work Schedule**

| Employee | Monday | Tuesday | Wednesday | Thursday | Friday |
|---|---|---|---|---|---|
| Roger | OFF | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | 7:30-5:30 |
| Lee | 7:30-5:30 | 7:30-5:30 | 7:30-5:30 | 7:30-5:00 | OFF |
| Betty | OFF | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | 7:30-5:30 |
| Greg | 7:30-5:30 | 7:30-5:30 | 7:30-5:30 | 7:30-5:00 | OFF |
| Leslie | OFF | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | 7:30-5:30 |
| Larry M | 7:30-5:30 | 7:30-5:30 | OFF | 7:30-5:00 | 7:30-5:30 |
| Tom | OFF | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | 7:30-5:30 |
| Larry G | 7:30-5:30 | 7:30-5:30 | 7:30-5:30 | 7:30-5:00 | OFF |
| Jeff | OFF | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | 7:30-5:30 |
| Aimee | 7:30-5:30 | 7:30-5:30 | 7:30-5:30 | 7:30-5:00 | OFF |
| Tad | 7:30-5:30 | 7:30-5:00 | 7:30-5:30 | OFF | 7:30-5:30 |

### TABLE 2

**Four-Day Workweek Statistics (Time expressed in hours)**
**FY 1994 = 12/1/93 - 11/31/94**

| Name | Mileage | Field Time | | Office Time | | Travel Time | | Total |
|---|---|---|---|---|---|---|---|---|
| | 1994 | 1994 | % | 1994 | % | 1994 | % | O/F/T |
| Jeff | 2,914 | 138 | 8% | 1,373 | 83% | 138 | 8% | 1,649 |
| Betty | 6,112 | 626 | 38% | 737 | 45% | 291 | 18% | 1,656 |
| Roger | 6,177 | 569 | 35% | 858 | 52% | 216 | 13% | 1,645 |
| Tom | 1,102 | 370 | 22% | 1,159 | 67% | 192 | 11% | 1,722 |
| Aimee | 3,281 | 324 | 20% | 1,172 | 71% | 155 | 9% | 1,653 |
| Greg | 4,997 | 270 | 16% | 1,234 | 73% | 182 | 11% | 1,687 |
| Larry G | 1,695 | 568 | 32% | 872 | 49% | 344 | 19% | 1,786 |
| Tad | 6,606 | 530 | 33% | 854 | 53% | 219 | 14% | 1,604 |
| Larry M | 1,063 | 333 | 19% | 1,167 | 68% | 223 | 13% | 1,724 |
| Leslie | 6,392 | 505 | 31% | 950 | 57% | 198 | 12% | 1,655 |
| Lee | 3,362 | 512 | 31% | 829 | 50% | 331 | 20% | 1,673 |
| **Total** | **43,701** | **4,748** | **26%** | **11,210** | **61%** | **2,495** | **14%** | **18,454** |

### TABLE 3

**Four-Day Workweek Statistics (Time expressed in hours)**
**FY 1993 = 12/1/92 - 11/31/93**

| Name | Mileage | Field Time | | Office Time | | Travel Time | | Total |
|---|---|---|---|---|---|---|---|---|
| | 1993 | 1993 | % | 1993 | % | 1993 | % | O/F/T |
| Jeff | 2,798 | 149 | 9% | 1,395 | 84% | 113 | 8% | 1,659 |
| Betty | 7,724 | 558 | 33% | 828 | 48% | 324 | 7% | 1,711 |
| Roger | 7,505 | 573 | 33% | 907 | 52% | 249 | 19% | 1,731 |
| Tom | 838 | 334 | 19% | 1,247 | 71% | 185 | 11% | 1,767 |
| Aimee | 4,592 | 353 | 21% | 1,192 | 69% | 176 | 10% | 1,722 |
| Greg | 8,107 | 534 | 31% | 953 | 55% | 254 | 15% | 1,743 |
| Larry G | 1,228 | 486 | 27% | 991 | 56% | 299 | 17% | 1,777 |
| Tad | 8,718 | 556 | 32% | 918 | 53% | 264 | 15% | 1,740 |
| Larry M | 1,097 | 227 | 13% | 1,357 | 76% | 195 | 11% | 1,780 |
| Leslie | 8,089 | 354 | 22% | 1,032 | 63% | 243 | 15% | 1,630 |
| Lee | 3,807 | 516 | 30% | 881 | 51% | 317 | 18% | 1,715 |
| **Total** | **54,503** | **4,645** | **24%** | **11,707** | **62%** | **2,622** | **14%** | **18,974** |

was expected to be at least five percent to 10 percent. The higher expectations were based upon the theory that with reductions in travel time, staff would spend longer hours out in the field performing their duties.

The second objective, to reduce employee automobile usage by 20 percent, was achieved. If each employee had driven to the office five days a week, there would have been a total of 55 trips into the office for those 11 employees. The 11 employees working a four-day week reduced their trips into the office to 44 trips, a 20 percent reduction. Fewer trips to the office also resulted in 20 percent fewer employee cars parked in the lot each week. As an added benefit, these fewer trips from home to the office kept more money in the employee's pockets by reducing their commuting expenses.

The EHS achieved the third objective of reducing mileage compensation. Staff travel was reduced to 10,802 miles, representing a savings of 20 percent (Table 4).

The fourth goal, to increase the public's and contractors' access to EHS by increasing the number of operating hours, was also achieved. The office hours were expanded from 42.5 hours per week to 50 hours per week, representing an increase of 18 percent.

The fifth objective of reducing use of sick leave and personal leave was not achieved. The use of personal leave and sick leave actually increased during the study period. By contrast, the use of annual vacation leave was reduced (Table 5).

The last objective, targeting reductions in child daycare costs, was achieved. Two employees were able to reduce the time their children spent in daycare.

## Discussion

Overall, the use of a four-day flexible work schedule has been successful at the EHS. Five out of the desired six objectives were achieved. Sanitarian field time was increased two percent during the evaluation period. Perhaps the low two percent increase was the result of procedural changes in how duties were being performed. Procedural changes in the food program could have resulted in the need for more office time. Perhaps with further emphasis on proper routing of travel, the expectation of future gains in field time will be realized.

Flexible working schedules have reduced auto use by 20 percent. These reductions have shown that employers can reduce auto use without compromising productivity. Employees working a four-day work week have reduced their personal travel expenses by driving one less day to the office.

The EHS has saved money in travel expenses paid to employees because of fewer trips back to the office by staff. During the evaluation period, there was a two percent decrease in total staff hours (compare total time from Tables 2 and 3). However, this small decrease in hours could not be responsible for nearly 11,000 less miles traveled by staff.

A natural gas company surveyed its customers and found the standard work day was a source of customer complaints. "By introducing flextime, the company was able to offer after-hour service to customers and provide its employees greater schedule flexibility without increased head count, overtime pay, or increased benefit costs" (3). EHS found similar gains after using flexible scheduling. The public and contractors have expressed satisfaction and have benefited from the increased access to services. Local contractors have expressed the greatest satisfaction because they can request inspections at 7:30 a.m. The extra hours have allowed contractors to make greater use of daylight hours, which increases their efficiency. Clients using EHS also have been appreciative of the increased office hours and added flexibility of early and late appointments. Many did not have to take off from work to use our services. A client survey needs to be developed fully to evaluate the benefit of longer hours.

Reduced use of sick leave and personal leave was not realized during the evaluation period. Perhaps a longer evaluation period would achieve that desired objective. The increase could have resulted from the stress of the longer hours. An employee survey should be developed to help answer this question. Work completion could have been an additional outcome measurement. However, the work load for each year is often dependent upon the economy. If interest rates are low and the economy is strong, the health department will have additional work loads that could affect the measurement of the effectiveness of the four-day work week.

A final feature expected was reduced child daycare costs. Two staff members with small children were able to reduce their daycare costs. They also expressed greater satisfaction of having more hours to spend personally raising their children. Benefits of decreased child care may be limited if five days per week contracts are required by the care givers.

EHS was able to achieve most of its objectives using the four-day flexible work schedule. Goals were achieved at a cost reduction to the agency. Since the test period was so successful, flexible hours were expanded to all Lake County Health Department employees.

Public organizations compete with the private sector to attract and retain trained, skilled

### TABLE 4

**Mileage Savings**

| Year | Total Miles Traveled |
|---|---|
| 1993 | 54,503 |
| 1994 | 43,701 |
| Savings | 10,802 |
| Percent Reduction | 20% |

### TABLE 5

**Four-Day Workweek Statistics (Time expressed in hours)**
**FY 1993 / FY 1994**

| Name | Annual Leave 1993 | Annual Leave 1994 | Personal Leave 1993 | Personal Leave 1994 | Sick Leave 1993 | Sick Leave 1994 |
|---|---|---|---|---|---|---|
| Jeff | 143 | 124 | 33 | 26 | 37 | 30 |
| Betty | 72 | 101 | 37 | 18 | 45 | 61 |
| Roger | 90 | 145 | 20 | 24 | 56 | 59 |
| Tom | 75 | 63 | 26 | 19 | 9 | 2 |
| Aimee | 150 | 84 | 1 | 17 | 92 | 79 |
| Greg | 105 | 93 | 2 | 34 | 65 | 63 |
| Larry G | 97 | 75 | 13 | 19 | 21 | 71 |
| Tad | 99 | 144 | 22 | 18 | 62 | 83 |
| Larry M | 112 | 66 | 22 | 9 | 26 | 35 |
| Leslie | 138 | 75 | 13 | 36 | 73 | 81 |
| Lee | 150 | 84 | 18 | 26 | 21 | 79 |
| **Total** | **1,223** | **1,057** | **211** | **247** | **511** | **647** |

and educated employees. The Lake County Health Department hopes that the added benefit of a three-day weekend and/or flexible work schedules will be an attractive lure.

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(Source: *Still Living Without the Basics, Rural Community Assistance Program*, 1995.)